

QUESTIONS 10 / 10 completed

Integer Array ▾

INTEGER ARRAY

Write a C program to demonstrate pointer arithmetic on an integer array. Display element values using both array indexing and pointer notation. Show address difference between consecutive elements.

Sample Input:

Enter number of

elements: 4

Enter elements:

10 20 30 40

Sample Output:

Using Array

Indexing:

arr[0] = 10

PROGRAM EDITOR

C ▾

Run

Save

```
1 #include<stdio.h>
2 int main()
3 {
4     int n,i,arr[n];
5     scanf("%d",&n);
6     int *ptr=arr;
7     for(i=0;i<n;i++)
8         scanf("%d",&arr[i]);
9     for(i=0;i<n;i++)
10        printf("arr[%d]=%d\n",i,arr[i]);
11    for(i=0;i<n;i++)
12        printf("***(ptr+%d)=%d\n",i,*(ptr+i));
13    printf("gap=%ld bytes \n",(long)(&arr[1])-(long)(&arr[0]));
14    return 0;
15 }
```

OUTPUT

```
arr[0]=10
arr[1]=20
arr[2]=30
arr[3]=40
arr[4]=-31459472
arr[5]=123
arr[6]=-31459472
arr[7]=123
arr[8]=-1
arr[9]=-1
```

INPUT

4
10 20 30 40

QUESTIONS 10 / 10 completed

Factorial

FACTORIAL

"Write a C program to find the factorial of a given number using recursion.
Sample Input: Enter a number- 6
Sample Output: Factorial of 6 is 720
"

PROGRAM EDITOR

C

Run

Save

```
1 #include<stdio.h>
2 int factorial(int n){
3     if (n==0||n==1){
4         return 1;
5     }
6     else{
7         return n*factorial(n-1);
8     }
9 }
10 int main(){
11     int num;
12     printf("enter a number:");
13     scanf("%d",&num);
14
15     printf("factorial of %d is %d\n",num,factorial(num));
16     return 0;
17 }
```

OUTPUT

```
enter a number:factorial of 6 is 720
```

INPUT

6

QUESTIONS 10 / 10 completed

Array Rotation

ARRAY ROTATION

"Implement array rotation (left/right) for arrays of size 10 by K positions without using extra space. Use a reversal algorithm. The program must validate K ($0 \leq K < N$), show original→rotated→reversal steps visually, compute the sum difference, and support multiple rotations. Sample Input/Output:
Enter 10 numbers:
1 2 3 4 5 6 7 8 9 10
Rotate Left

PROGRAM EDITOR

C

Run

Save

```
1 #include<stdio.h>
2 int main()
3 {
4     int a[5]={1,2,3,4,5};
5     int i,temp;
6
7     temp=a[0];
8
9     for (i=0;i<4;i++){
10         a[i]=a[i+1];
11     }
12     a[4]=temp;
13
14     printf("array after rotation:\n");
15     for(i=0;i<5;i++)
16     {
17         printf("%d",a[i]);
18     }
19
20     return 0;
21 }
```

OUTPUT

```
array after rotation:
23451
```

INPUT

Your INPUT goes here!

QUESTIONS 10 / 10 completed

Sparse Matrices

SPARSE MATRICES

Represent two sparse matrices (mostly zeros) using arrays of triples [row, col, value]. Read dimensions and non-zero terms for two matrices. Add them, store result in sparse form (sorted by row then col), and display dense matrix form. Max 10x10; ignore zeros.

Sample Input
(Matrix 1):
3 3 3
0 1 5
1 0 8

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2
3 int main()
4 {
5     int a[3][3], i, j, count=0;
6     printf("enter matrix:\n");
7
8     for (i=0; i<3; i++){
9         for (j=0; j<3; j++){
10             scanf("%d", &a[i][j]);
11
12             if(a[i][j]!=0)
13                 count++;
14         }
15     }
16     if(count>4)
17         printf("sparse matrix");
18     else
19         printf("not a sparse matrix");
20     return 0;
21 }
```

OUTPUT

```
enter matrix:
not a sparse matrix
```

INPUT

Your INPUT goes here!

QUESTIONS 10 / 10 completed

Fibonacci Series ▾

FIBONACCI SERIES

Write a C program to print the first n terms of the Fibonacci series ($n \leq 20$) without using recursion. Start with 0 and 1.

Sample Input/Output:

Enter number of terms: 8

Fibonacci series: 0 1 1 2

3 5 8 13

PROGRAM EDITOR

C ▾

Run

Save

```
1 #include<stdio.h>
2
3 int main(){
4     int n,a=0,b=1,c,i;
5     printf("enter number of terms:");
6     scanf("%d",&n);
7     printf("Fibonacci series:");
8     for(i=1;i<=n;i++){
9         printf("%d",a);
10        c=a+b;
11        a=b;
12        b=c;
13    }
14    return 0;
15 }
```

OUTPUT

```
enter number of terms:Fibonacci series:01123
```

INPUT

5

QUESTIONS 10 / 10 completed

Array

ARRAY

Write a C program to find the 5th element in the given array/

Sample Input: a=[10,20,30,40,50,60]
Sample output: 50

PROGRAM EDITOR

C

Run

Save

```
1 #include<stdio.h>
2 int main(){
3     int a[10],i;
4
5     printf("enter 10 elements:\n");
6     for(i=0;i<10;i++){
7         scanf("%d",&a[i]);
8     }
9     printf("the 5th elements is:%d",a[4]);
10    return 0;
11 }
```

OUTPUT

```
enter 10 elements:
the 5th elements is:50
```

INPUT

```
10 20 30 40 50 60 70
80 90 100
```

QUESTIONS 10 / 10 completed

Fibonacci series (N

FIBONACCI SERIES (N ≤ 50)

Develop a C program to print the first N terms of the Fibonacci series (N ≤ 50) without using recursion. Use two variables to generate the series iteratively and store the series in an array. The program should also calculate and display the sum of the series and identify the largest term. Input validation is mandatory for N (1 ≤ N ≤ 50).

PROGRAM EDITOR

C

Run

Save

```
1 #include<stdio.h>
2 int main()
3 {
4     int n, a = 0, b = 1, c , i;
5     int f[50], sum=0;
6
7     printf("enter number of terms (1-50):");
8     scanf("%d",&n);
9     if(n<1||n>50){
10         printf("invalid input!please enter terms between 1 and 50:");
11         return 0;
12     }
13     printf("fibonacci series:");
14
15     for (i=0;i<n;i++){
16         printf("%d",a);
17         f[i]=a;
18         sum=sum+a;
19
20         c=a+b;
21         a=b;
22         b=c;
23     }
```

OUTPUT

```
enter number of terms (1-50):fibonacci series:0112358132134
sum of series:88
largest term:34
```

INPUT

10

QUESTIONS 10 / 10 completed

Unsorted Array ▾

UNSORTED ARRAY

Write a program to find the missing element for the given unsorted array of sequence numbers.

Sample Input: a=

[10,9,7,4,3,1,2]

Sample output: 5,6,8

PROGRAM EDITOR

C ▾

Run

Save

```
1 #include<stdio.h>
2 int main()
3 {
4     int a[]={10,9,7,4,3,1,2};
5     int i,j,found;
6
7     for(i=1;i<=10;i++)
8     {
9         found=0;
10        for(j=0;j<7;j++)
11        {
12            if(a[j]==i)
13            {
14                found=1;
15                break;
16            }
17        }
18        if(found==0)
19            printf("%d",i);
20
21    }
```

OUTPUT

568

INPUT

Your INPUT goes here!

QUESTIONS 10 / 10 completed

Fib series-iterative ▾

**FIB SERIES-
ITERATIVE &
RECUR**

Write a C program to generate Fibonacci series up to N terms using both iterative and recursive methods. The program should display both outputs and compare execution time in milliseconds. Also display total recursive calls made.

Sample Input:

Enter N: 7

Sample Output:

Iterative: 0 1 1 2 3 5 8

Recursive: 0 1 1 2 3 5 8

Recursive Calls: 13

Iterative Time: 0.001 ms

Recursive Time: 0.005

ms

PROGRAM EDITOR

C ▾

Run

Save

```
1 #include<stdio.h>
2 int fib(int n)
3 {
4     if(n==0)
5         return 0;
6     if(n==1)
7         return 1;
8     return fib(n-1)+fib(n-2);
9 }
10 int main()
11 {
12     int n,i,a=0,b=1,c;
13     printf("enter N:");
14     scanf("%d",&n);
15
16     printf("iterative:");
17     for(i=0;i<n;i++)
18     {
19         printf("%d,a");
20         c=a+b;
21         a=b;
```

OUTPUT

```
enter N: iterative: 0, a0, a1, a1, a2, a3, a5, a
Recursive: 0112358
```

INPUT

7

QUESTIONS 10 / 10 completed

Structure ▾

STRUCTURE

Write a C program to store student details (Roll No, Name, Marks) using structure and save them into a file. Display highest mark, class average, and total records stored.

Sample Input:
Enter number of students: 2
101 Ravi 75
102 Kiran 85
Sample Output:
Highest Mark: 85 (Kiran)
Class Average: 80.00
2 records stored successfully.

Test Case:

PROGRAM EDITOR

C ▾

Run

Save

```
1 #include<stdio.h>
2 struct student
3 {
4     int roll,marks;
5     char name[20];
6
7 };
8 int main()
9 {
10     struct student s[10];
11     int n,i,total=0,max=0,p=0;
12     scanf("%d",&n);
13     for(i=0;i<n;i++)
14     {
15         scanf("%d%s%d",&s[i].roll,s[i].name,&s[i].marks);
16         total+=s[i].marks;
17         if(s[i].marks>max)
18             max=s[i].marks;
19     }
20     printf("highest marks:%d(%s)\n",max,s[p].name);
21     printf("average:%.2f\n",(float)total/n);
```

OUTPUT

```
highest marks:85(kiran)
average:80.00
2 records stored
```

INPUT

```
2
101 ravi 75
102 kiran 85
```